

*Heaven's light is our guide*

## Rajshahi University of Engineering & Technology



Department of Electrical & Computer Engineering

**Course No. : ECE 2112**

**Course Title : Digital Techniques Sessional**

**Lab Report : 02**

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| <b><u>Submitted by:</u></b><br>Rafin Mahfuz<br>Roll : 2410029<br>Department of Electrical<br>& Computer Engineering,<br>RUET | <b><u>Submitted to:</u></b><br>Mst. Mazeda Noor Tasnim<br>Lecturer<br>Department of Electrical<br>& Computer Engineering,<br>RUET |
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**Date of Submission : 16 August 2026**

## Problem No: 01

**Problem Statement:** Simplifying the Boolean expression  $F(A, B, C) = A'BC + AB'C + ABC' + ABC$  using Boolean algebra and a Karnaugh map, and then checking the original and simplified circuits through logic simulation.

**Objective:** To simplify a three-variable Boolean expression using Boolean algebra and a Karnaugh map, and then verify that the original and simplified circuits give the same output using Logisim.

### Algebraic Simplification:

$$\begin{aligned}
 F(A, B, C) &= A'BC + AB'C + ABC' + ABC \\
 &= A'BC + ABC + AB'C + ABC + ABC' + ABC \\
 &= BC(A' + A) + AC(B' + B) + AB(C' + C) \\
 &= BC + AC + AB \\
 &= AB + AC + BC
 \end{aligned}$$

### Truth Table:

| A | B | C | $F = AB + AC + BC$ |
|---|---|---|--------------------|
| 0 | 0 | 0 | 0                  |
| 0 | 0 | 1 | 0                  |
| 0 | 1 | 0 | 0                  |
| 0 | 1 | 1 | 1                  |
| 1 | 0 | 0 | 0                  |
| 1 | 0 | 1 | 1                  |
| 1 | 1 | 0 | 1                  |
| 1 | 1 | 1 | 1                  |

### Karnaugh Map:

| A \ BC | BC |    |    |    |
|--------|----|----|----|----|
|        | 00 | 01 | 11 | 10 |
| 0      | 0  | 0  | 1  | 0  |
| 1      | 0  | 1  | 1  | 1  |

This is a 3-variable K-map for A, B, and C. The 1s show where the output is 1. By grouping the 1s, the simplified expression is:

$$F = AB + AC + BC$$

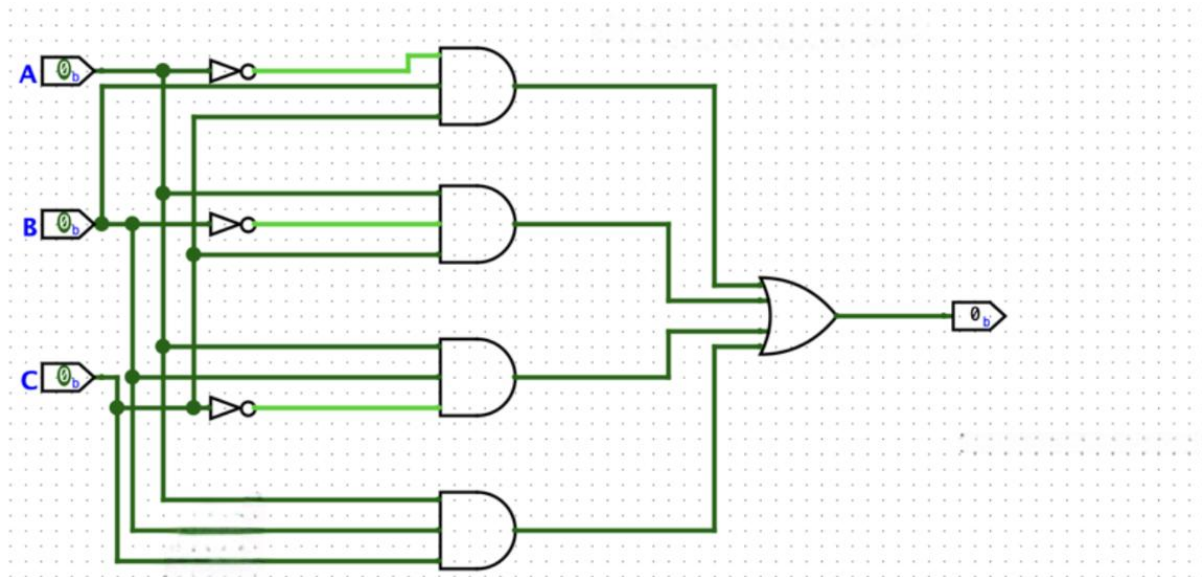
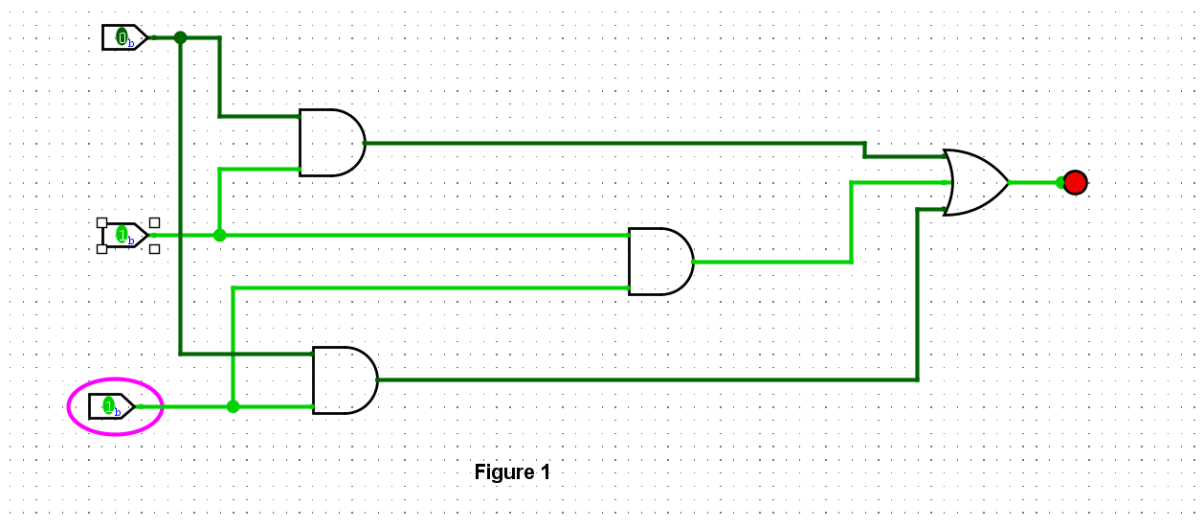
**Circuit Diagram:****Original Circuit:****Simplified Circuit:**

Figure 1

**Discussion:**

In this experiment, the given Boolean expression was simplified using both Boolean algebra and the Karnaugh map (K-map) method. Both methods produced the same simplified expression,  $F = AB + AC + BC$ . The original and simplified circuits were then implemented and tested in Logisim. The simulation results showed that both circuits produced the same output for all possible input combinations. Therefore, the simplified circuit is functionally equivalent to the original circuit and uses fewer logic gates.

## Problem No: 02

**Problem Statement:** Simplifying the Boolean expression  $F(A,B,C)=A(A+B)(A+B+C)$  using Boolean algebra and a Karnaugh map, and then checking the original and simplified circuits using logic simulation.

**Objective:** To simplify a three-variable Boolean expression using Boolean algebra and a Karnaugh map, and verify that the original and simplified circuits give the same output using Logisim.

**Algebraic Simplification:**

$$\begin{aligned}
 F(A,B,C) &= A(A+B)(A+B+C) \\
 &= (A+AB)(A+B+C) \\
 &= A(A+B+C) + AB(A+B+C) \\
 &= A + AB + AC + AB + AB + ABC \\
 &= A + AB + AC + ABC \\
 &= A(1 + B + C + BC) \\
 &= A
 \end{aligned}$$

**Truth Table:**

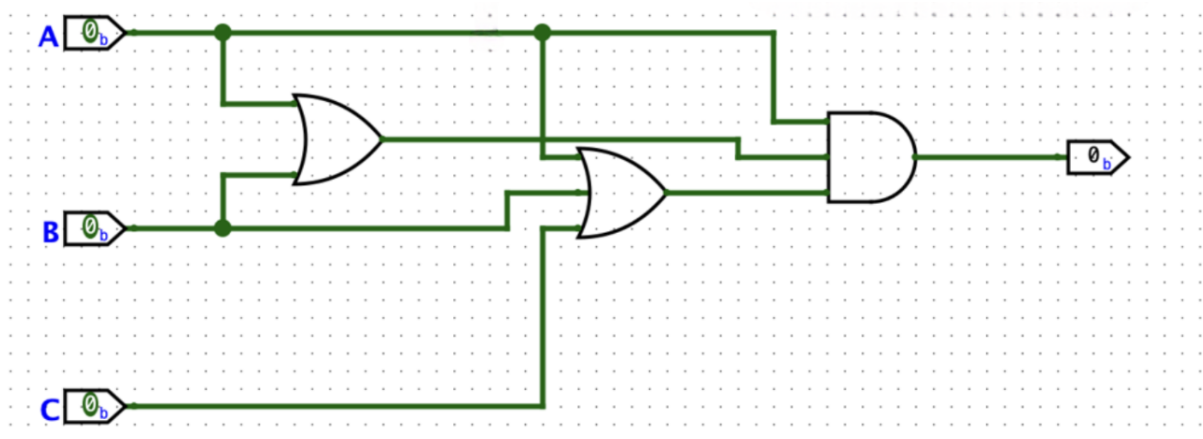
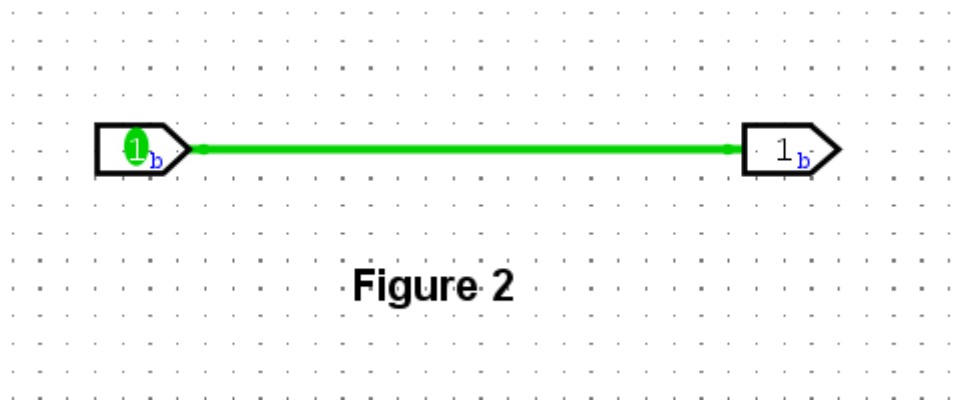
| A | B | C | F = A |
|---|---|---|-------|
| 0 | 0 | 0 | 0     |
| 0 | 0 | 1 | 0     |
| 0 | 1 | 0 | 0     |
| 0 | 1 | 1 | 0     |
| 1 | 0 | 0 | 1     |
| 1 | 0 | 1 | 1     |
| 1 | 1 | 0 | 1     |
| 1 | 1 | 1 | 1     |

**Karnaugh Map:**

|   |   | BC |    |    |    |
|---|---|----|----|----|----|
|   |   | 00 | 01 | 11 | 10 |
| A | 0 | 0  | 0  | 0  | 0  |
|   | 1 | 1  | 1  | 1  | 1  |

This is a 3-variable K-map for A, B and C. The entire row where  $A = 1$  contains 1s, while the row where  $A = 0$  contains 0s. By grouping the four 1s together, B and C are eliminated. Therefore, the simplified expression is:

$$F = A$$

**Circuit Diagram:****Original Circuit:****Simplified Circuit:****Figure 2****Discussion:**

In this experiment, the given Boolean expression was simplified using Boolean algebra and the Karnaugh map method. Both methods gave the same simplified expression,  $F = A$ . The original and simplified circuits were then implemented and tested in Logisim. The simulation showed that both circuits produced the same output for all possible input combinations. This confirms that B and C do not affect the output.

### Problem No: 03

**Problem Statement:** Simplifying the Boolean expression  $F(A, B, C) = (A + (BC)')'(AB + ABC)$  using Boolean algebra and a Karnaugh map, and then checking the original and simplified circuits using logic simulation.

**Objective:** To simplify a Boolean expression using De Morgan's theorem and Boolean algebra, verify the result using a Karnaugh map, and check the original and simplified circuits using Logisim.

**Algebraic Simplification:**

$$\begin{aligned}
 F(A, B, C) &= (A + (BC)')'(AB + ABC) \\
 &= A' \cdot BC \cdot (AB + ABC) \quad (\text{De Morgan's theorem}) \\
 &= A'BC \cdot AB(1 + C) \\
 &= A'BC \cdot AB \\
 &= (AA')BCB \\
 &= 0
 \end{aligned}$$

Therefore, the simplified expression is:

$$F = 0$$

**Truth Table:**

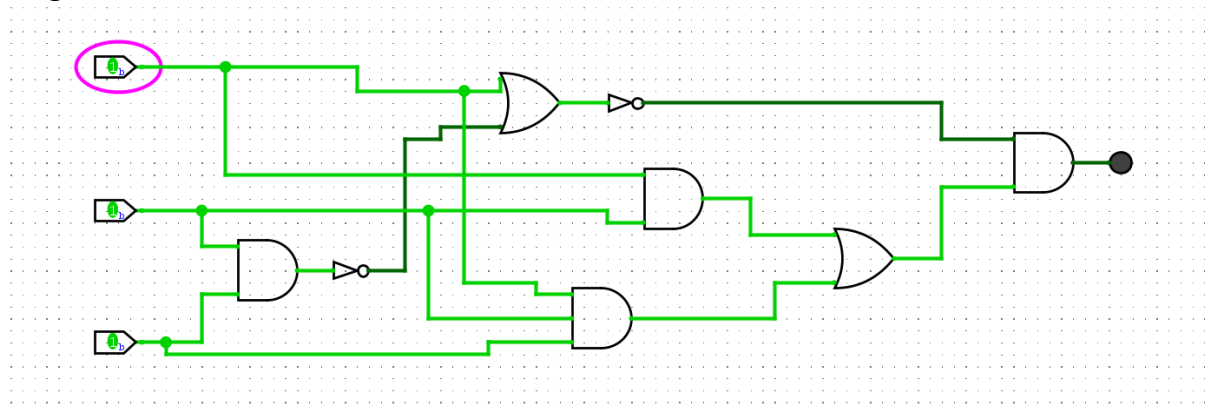
| A | B | C | F = 0 |
|---|---|---|-------|
| 0 | 0 | 0 | 0     |
| 0 | 0 | 1 | 0     |
| 0 | 1 | 0 | 0     |
| 0 | 1 | 1 | 0     |
| 1 | 0 | 0 | 0     |
| 1 | 0 | 1 | 0     |
| 1 | 1 | 0 | 0     |
| 1 | 1 | 1 | 0     |

**Karnaugh Map:**

| A \ BC | BC |    |    |    |
|--------|----|----|----|----|
|        | 00 | 01 | 11 | 10 |
| 0      | 0  | 0  | 0  | 0  |
| 1      | 0  | 0  | 0  | 0  |

This is a 3-variable K-map for A, B, and C. All the cells contain 0, so there are no 1s to group. Therefore, the simplified expression is:

$$F = 0$$

**Circuit Diagram:****Original Circuit:****Simplified Circuit:****Figure 3****Discussion:**

In this experiment, the given Boolean expression was simplified using De Morgan's theorem and Boolean algebra. The expression was reduced to  $F = 0$  because the term  $AA'$  is always 0. The Karnaugh map also contained only 0s, confirming the same result. Both the original and simplified circuits were tested in Logisim, and the output remained 0 for all input combinations. Therefore, both circuits are functionally equivalent.

### Problem No: 04

**Problem Statement:** Simplifying the Boolean expression  $F(A, B) = (B'(A + B) + (A + B)(A + B'))B'$  using Boolean algebra and a Karnaugh map, and then checking the original and simplified circuits using logic simulation.

**Objective:** To simplify a two-variable Boolean expression using Boolean algebra and a Karnaugh map, and verify that the original and simplified circuits give the same output using Logisim.

**Algebraic Simplification:**

$$\begin{aligned}
 F(A, B) &= (B'(A + B) + (A + B)(A + B'))B' \\
 &= (AB' + BB' + AA + AB' + AB' + BB')B' \\
 &= (A + AB' + AB' + AB')B' \\
 &= A(1 + B' + B' + B')B' \\
 &= AB'
 \end{aligned}$$

Therefore, the simplified expression is:

$$F = AB'$$

**Truth Table:**

| A | B | $F = AB'$ |
|---|---|-----------|
| 0 | 0 | 0         |
| 0 | 1 | 0         |
| 1 | 0 | 1         |
| 1 | 1 | 0         |

**Karnaugh Map:**

|                  |   |   |
|------------------|---|---|
| $A \backslash B$ | 0 | 1 |
|                  | 0 | 0 |
| 1                | 1 | 0 |

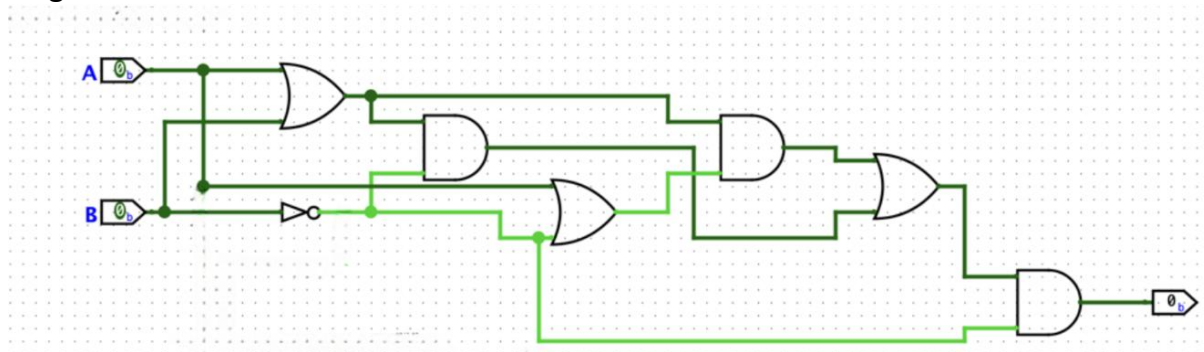
This is a 2-variable K-map for A and B. Only one cell contains 1, which occurs when A = 1 and B = 0. Therefore, the simplified expression is:

$$F = AB'$$

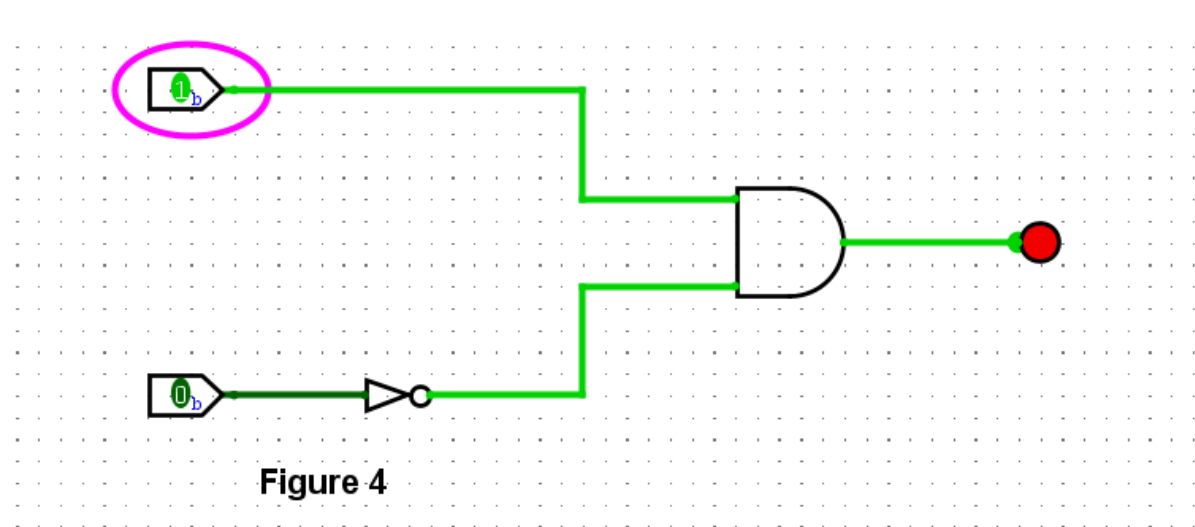


### Circuit Diagram:

**Original Circuit:**



### Simplified Circuit:



### Figure 4

### Discussion:

In this experiment, the given Boolean expression was simplified using Boolean algebra and the Karnaugh map method. Both methods produced the same simplified expression,  $F = AB'$ . The original and simplified circuits were then implemented and tested in Logisim. The simulation showed that the output is 1 only when  $A = 1$  and  $B = 0$ . For all other input combinations, the output is 0. Therefore, both circuits are functionally equivalent.

## Conclusion:

In this experiment, four different Boolean expressions were successfully simplified using Boolean algebra and Karnaugh maps. The simplified results were obtained as  $F = AB + AC + BC$ ,  $F = A$ ,  $F = 0$ , and  $F = AB'$ , respectively. The Karnaugh maps confirmed the results obtained through algebraic simplification. The original and simplified circuits were also implemented and tested in Logisim, and the simulation results showed identical outputs for all possible input combinations. These results verify that Boolean simplification can significantly reduce circuit complexity, minimize the number of logic gates, and produce more efficient and practical digital circuits without changing their functionality.